# Technical debt review and recommendations

Date: 12 November 2026
Reviewer role: Programme management / technical debt
Release under review: 2.7.0 (development 172.16.16.44)
Method: file-size and hotspot inventory, test/CI review, incident review of the 12 November availability stall, security backlog, dependency review, and the data-quality audit.

## Executive summary

The platform is feature-rich and the delivery discipline is above average: supply-chain locking, lab validation, CSP hashing, documentation builds, and a large backend test suite are already in place. The debt is concentrated, not spread: four very large files hold most of the complexity, the frontend has no unit or E2E gate in CI, lint is non-blocking, and the single-worker backend has no guardrail against blocking work on the event loop.

The 12 November outage was not a feature bug. A lab spawn/teardown held a database session while the instructor dashboard fired a burst of analytics requests. The 4+4 pool exhausted, the next auth lookup blocked the event loop for 30 seconds, and every page appeared to hang. The immediate fixes are in place; the architectural pattern that allowed it is the top debt item.

## Debt inventory

Area | Evidence | Risk | Effort | Priority
Monolithic admin UI | Admin.vue 14,389 lines; InstructorPanel.vue 4,626; TrackDetail.vue 4,432 | Review cost, merge conflicts, regression risk | High | P1
Monolithic backend routers | admin.py 4,496 lines; docker_manager.py 4,048; courses.py 2,657; analytics.py 2,382 | Hard to test in isolation; slow review | High | P1
Event-loop discipline | Sync DB/Docker work in async endpoints; pool exhaustion incident | Total platform stall | Medium | P1
Frontend test gap | No unit tests, no Playwright job in the default CI; ui-smoke.yml needs credentials | Button/flow regressions reach users | Medium | P1
Lint gate | 51 ruff findings; CI continue-on-error: true; 132 ESLint warnings | Drift and dead code accumulate | Low | P2
N+1 queries | 91 relationships, few eager loads; analytics/dashboard routers | Latency at scale | Medium | P2
Retention | Ledger/activity/flag_attempts grow without a prune job | Storage and review-queue noise | Low | P2
Dependency currency | Postgres 15 runtime vs 17 CI; Python 3.11; base images tag-pinned | Upgrade crunch, supply-chain drift | Medium | P2
Auth storage | JWT in localStorage (mitigated by CSP + DOMPurify) | XSS token theft if CSP regresses | High | P2
Deferred hardening | RangeBox pinning/supervisord; Valkey requirepass; fail-closed revocation; username timing | Security depth | Medium | P2
Content scale | 169 labs, 160 workbook pages, 43 chapters; many review_status pending live walkthrough | Content quality drift | Medium | P3
Repo sprawl | opencyberrange, -ha, -next, -v1.1-fix, backups on the workstation | Wrong-tree edits, stale docs | Low | P3

## Top recommendations

### 1. Split Admin.vue and admin.py by domain (P1)

Extract the System, Users, Courses, Events, Achievements, and Question Bank panels into components with their own props/events. On the backend, split

admin.py into admin_users.py, admin_system.py, admin_content.py, and admin_reports.py behind the same router prefix. Acceptance: no file above 1,500 lines; routes unchanged; tests pass.

### 2. Make the event loop non-blocking by rule (P1)

Adopt one rule: no synchronous DB, Docker, filesystem, or report generation in an async def endpoint. Move work to asyncio.to_thread/run_in_executor, or make the endpoint def so FastAPI uses the threadpool. Add a CI check that flags SessionLocal() or docker_manager. calls inside async def bodies. Acceptance: the pool-burst test and the scheduler tick stay under 50 ms p95.

### 3. Add a real frontend gate (P1)

Wire a Playwright job into CI with a seeded test database: student, instructor, and admin login; dashboard, exercises, profile, admin System, question banks, achievements, events, and the CyberChef panel. Keep ui-smoke.mjs and extend it rather than creating a second harness. Acceptance: the job is required and runs in under 10 minutes.

### 4. Turn lint into a ratchet (P2)

Fix the 51 ruff findings (36 auto-fixable) and remove continue-on-error from the ruff step. For ESLint, fail on errors now and track warnings with a budget that must not increase. Acceptance: zero ruff errors; warning count frozen.

### 5. Eager-load the hot routers (P2)

Audit analytics.py, dashboard.py, admin.py, and courses.py for loops over relationships. Use selectinload for collections and joinedload for many-to-one; use count() where rows are not needed. Gate with the existing stress test. Acceptance: p95 latency and query count improve on the roster and dashboard scenarios.

### 6. Schedule retention (P2)

Add a scheduler job for data_integrity_ledger, activity_events, and flag_attempts, with a documented retention window and a run record in the ledger. Acceptance: row counts stabilize; the data analyst review queue is smaller.

### 7. Separate report signing and rotate secrets (P2)

Set REPORT_SIGNING_KEY independently of JWT_SECRET; document a rotation runbook for WG_ENCRYPTION_KEY and REPORT_SIGNING_KEY. Acceptance: a rotation drill completes without losing access to stored secrets.

### 8. Finish the deferred security depth (P2)

RangeBox supervisord + digest pinning, Valkey requirepass, fail-closed revocation for privileged roles, username-enumeration timing, and digest pinning for base images. Acceptance: each item has a test or a recorded accepted risk.

### 9. Upgrade with a rehearsal (P2)

Postgres 15 -> 17 and Python 3.11 -> 3.12, one at a time, on a restored copy. Record timings and rollback steps. Acceptance: a staging restore and upgrade complete inside the maintenance window.

### 10. Keep the release gate honest (P1 process)

Use docs/PROD-DEPLOYMENT-VALIDATION-MATRIX.md for every promotion. A release does not ship with NOT RUN rows. Acceptance: 100% required rows PASS and signed.

## Guardrails to stop regrowth

1. One PR per numbered item; no drive-by features.
2. A stress-test baseline diff for anything on a request path.
3. No new inline migrations in main.py; Alembic only.
4. Every new endpoint has a role test and a rate-limit decision.
5. Every new UI surface has a loading, empty, error, and permission-denied state.
6. The production matrix is the only release gate; chat approvals do not count.

## What not to do

- Do not rewrite the frontend or adopt a new framework to solve file size.
- Do not add a second theme system or a second test harness.
- Do not raise the DB pool again without fixing long-held sessions.
- Do not promote to production while restore, upgrade, and rollback drills are NOT RUN.